

SUPERMARKET SALES ANALYSIS

Professional Data Analysis Report

Executive Summary: The dataset contains 9,994 transaction rows covering January 2014 to December 2017. Total sales were \$2,297,200.86, total profit was \$286,397.02, and the average order value was \$458.61. The analysis identifies category, regional, temporal and discount-related patterns while preserving loss-making transactions for investigation.

1. Dataset & Data Quality

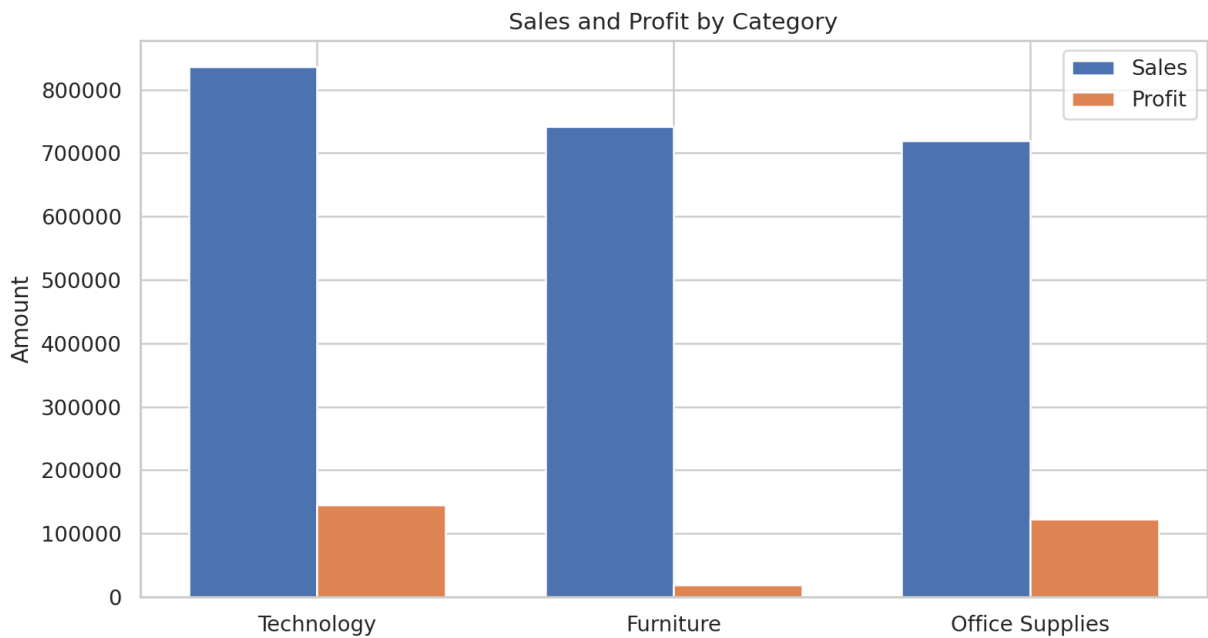
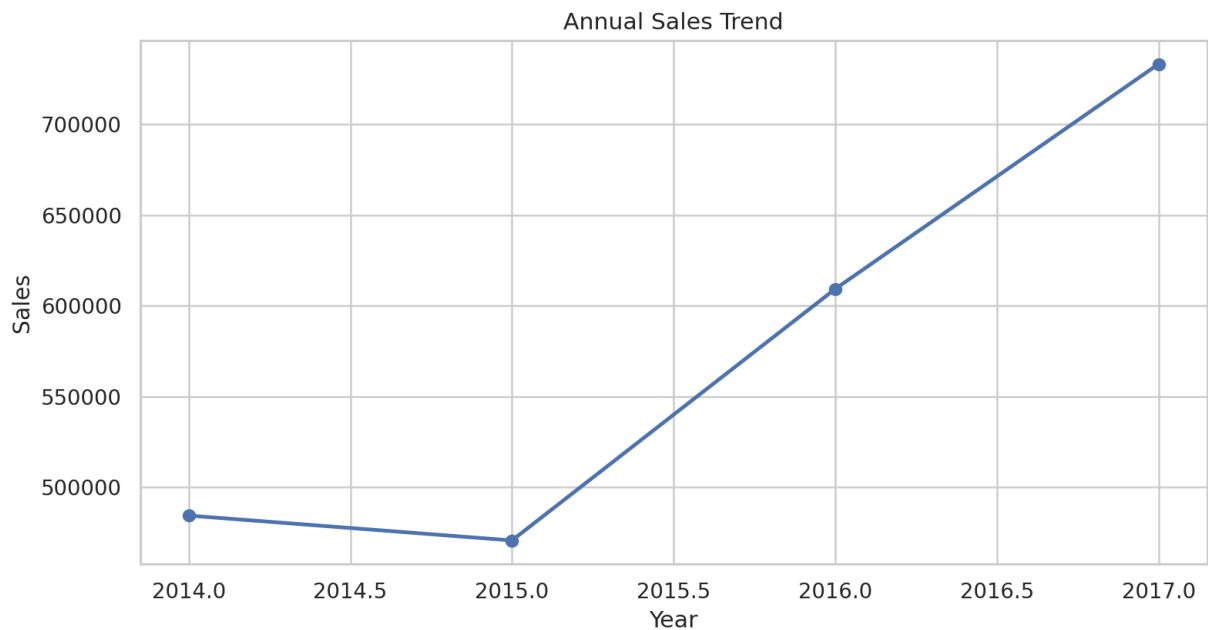
The dataset contains customer, order, product, geographic, sales, quantity, discount and profit fields. There are no missing cells and no duplicate rows. Dates were converted to datetime and derived fields were created for year, month, weekday and shipping duration.

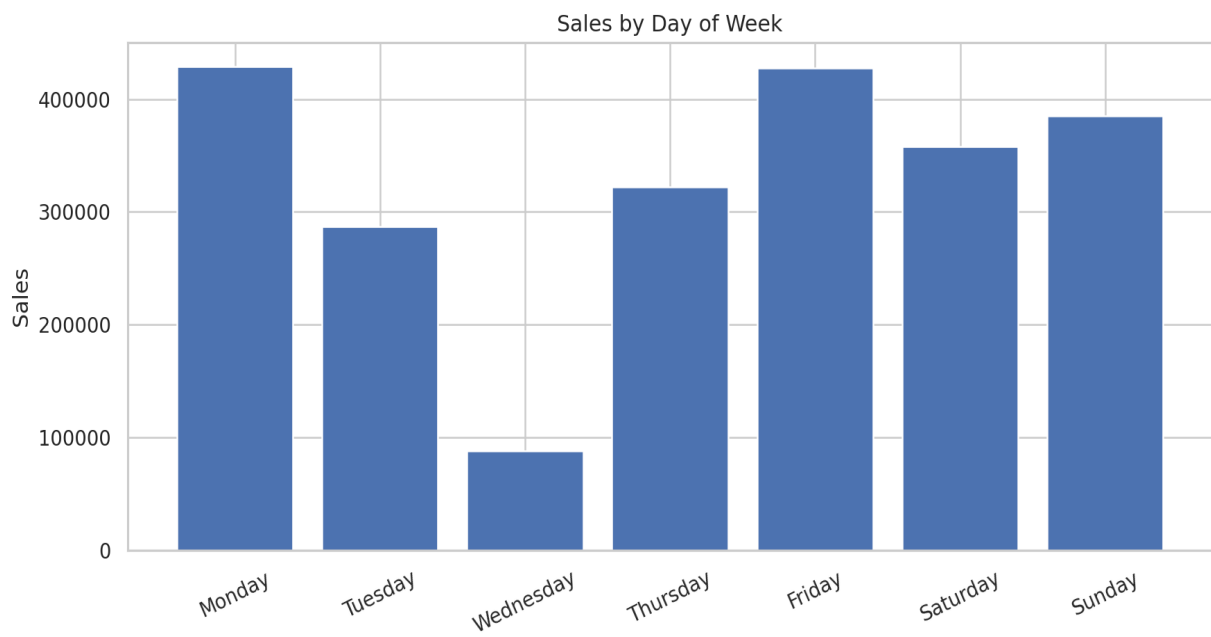
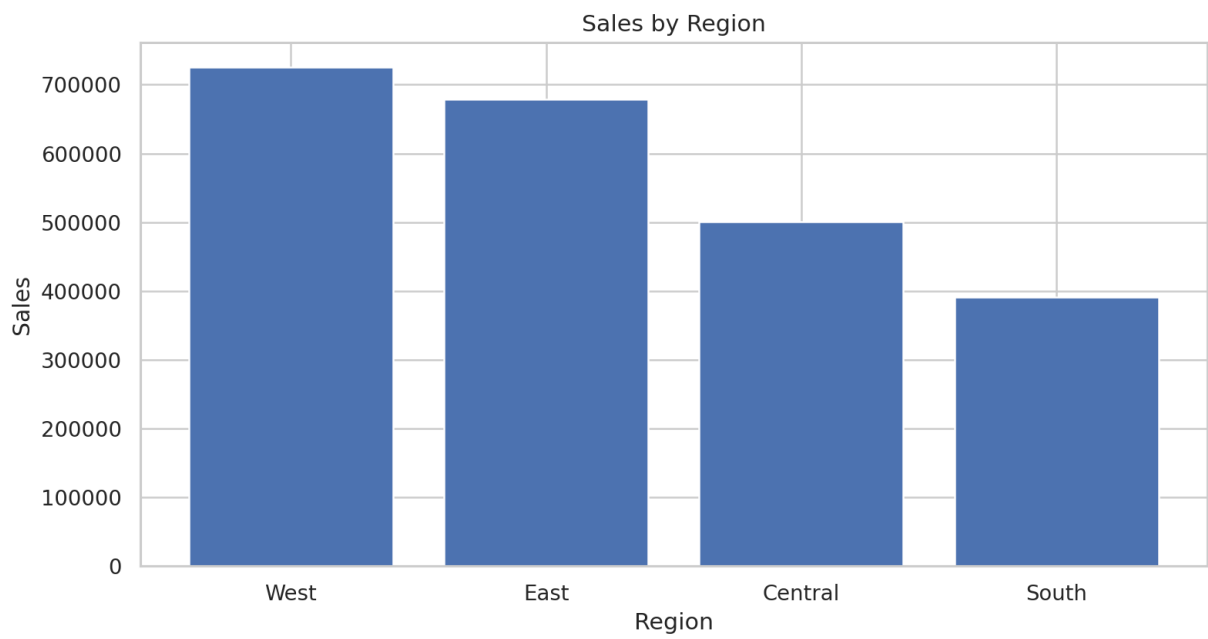
Metric	Value
Rows	9,994
Unique Orders	5,009
Unique Customers	793
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Total Quantity	37,873
Average Order Value	\$458.61
Profit Margin	12.47%

2. Key Findings

- Technology is the highest-sales category at \$836,154, while Office Supplies has the highest quantity sold (22,906 units).
- The West region records the highest sales at \$725,458.
- Annual sales increased from \$484,247 in 2014 to \$733,215 in 2017.
- Tables have high sales (\$206,966) but negative aggregate profit (\$-17,725), indicating a margin-management issue worth investigating.
- There are 1,871 loss-making transaction rows. This makes profit-focused analysis important rather than relying on sales alone.
- Discount and profit have a correlation of -0.219; this is an association and does not by itself prove that discounts cause lower profit.

3. Visual Analysis





4. Recommendations

- Review pricing, discounting and product-level economics for loss-making sub-categories, particularly Tables and Bookcases.
- Study the operating and product mix factors behind strong West-region sales before attempting to replicate them in other regions.
- Track gross profit and margin alongside revenue when evaluating products and promotions.
- Use customer segment and product-level discount analysis before changing promotion rules.
- Automate this analysis on a recurring schedule so management can monitor changes in sales, profit and loss-making products.

5. Limitations

The dataset does not include inventory, acquisition cost, marketing spend or customer satisfaction. Therefore, the analysis can identify patterns and associations but cannot establish causal effects or fully explain profitability. Findings are specific to the supplied dataset and its 2014–2017 period.